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Track: Cyber Security
Task: Task 1 - Introduction to Cybersecurity

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1. Cybersecurity Notes

What is Cybersecurity?
Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks. These cyberattacks are usually aimed at accessing, changing, or destroying sensitive information; extorting money from users; or interrupting normal business processes.

The CIA Triad (Core Principles):

- **Confidentiality:** Ensuring that data is only accessed by authorized people. (e.g., using strong passwords and encryption).
- **Integrity:** Ensuring that data is accurate and has not been tampered with or altered by unauthorized parties.
- **Availability:** Ensuring that systems and data are accessible to authorized users whenever they need them (e.g., preventing server crashes or DDoS attacks).

Personal vs. Organizational Security Risks:

- **Personal Risks:** Identity theft, loss of personal photos/files, financial loss (bank hacking), and social media account hijacking.
- **Organizational Risks:** Data breaches (leaking customer data), financial loss due to ransomware, damage to company reputation, and legal penalties for not protecting data.

2. List of 10 Common Cyber Threats

1. **Phishing:** Sending fake emails or messages that look legitimate to trick people into revealing sensitive information like passwords or credit card numbers.
2. **Malware:** Malicious software (like viruses, worms, and trojans) designed to damage or gain unauthorized access to a computer system.
3. **Ransomware:** A type of malware that locks your files and demands payment (ransom) to unlock them.
4. **Password Attacks:** Attempting to crack passwords through methods like brute force (guessing repeatedly) or dictionary attacks.
5. **Man-in-the-Middle (MitM) Attack:** When a hacker secretly intercepts and relays communication between two parties who believe they are directly communicating with each other.

6. **Denial-of-Service (DoS):** Flooding a system or network with traffic to make it crash or become unavailable to legitimate users.
7. **Social Engineering:** Manipulating people psychologically to trick them into making security mistakes or giving away sensitive information.
8. **SQL Injection:** Inserting malicious code into a website's database to access, modify, or delete private data.
9. **Zero-Day Exploit:** A cyberattack that occurs on the same day a weakness is discovered in software, before a fix is available.
10. **Insider Threat:** A security risk that originates from within the targeted organization, such as a disgruntled employee stealing data.